



Tertiary Infotech Academy Pte Ltd

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LAB 04 INSTRUCTIONS

For

Color and annotate evidence

Course Code: C1123

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Version 5.1

Lab 04 — Color and annotate evidence

Course: Wireshark Network Analysis Masterclass (C1123)

Topic: 04 — Navigation and Coloring Techniques

Learning outcome: LO1 — Plan scoped captures and configure a reproducible analyst profile.

Time: about 75 minutes

Version: v5.1 · 5 October 2026

Scenario

Users see intermittent web errors. You need the failing responses to stand out, and annotations that travel with the evidence.

Goal

Color rules are applied in order; the first matching rule wins.

What you will produce

An annotated pcapng with a comment on the failing response.

Before you start

- Wireshark 4.6 or later, with the TShark command-line tools on PATH.
- Python 3 for the verification and export scripts (Windows: use `py -3` in place of `python3`).
- Open this lab folder in a terminal; every file the lab needs is inside it.
- Use only the supplied synthetic captures — do not capture on a network you are not authorised to monitor.

Files for this lab

File	What it is for
data/branch-office.pcap	Synthetic branch-office capture used by the lab
assets/scenario.md	The help-desk ticket that sets the scenario
assets/checks.json	Filters and frame counts the fixture must satisfy
assets/expected-evidence.png	The packet list your filter should produce
scripts/verify.py	Checks the capture facts with TShark
scripts/export_evidence.py	Exports this lab's evidence rows to outputs/evidence.csv
outputs/findings.md	Findings sheet you complete

Lab at a glance

1. Filter HTTP 4xx and 5xx responses
2. Add a LAB HTTP Error colouring rule
3. Mark and comment the 500 response
4. Save and reopen the annotated pcapng

Step-by-step

1. Prepare your lab workspace. Open this lab folder in a terminal. Windows uses `py -3` in place of `python3`. Wireshark GUI alone is sufficient for the investigation; CLI scripts require TShark on PATH. Run the fixture verification first.

```
python3 scripts/verify.py
```

2. Open `data/branch-office.pcap`. Apply `http.response.code >= 400`. Record the 404 and 500 frame numbers.

3. Open View > Coloring Rules. Add LAB HTTP Error with expression `http.response.code >= 400`, dark text and a pale amber background. Move it above a general HTTP rule.

4. Select the 500 response, mark it using Edit > Mark/Unmark Packet and add a packet comment describing the observed status only. Save As `outputs/annotated.pcapng` to preserve comments.

5. Clear the filter. Use Edit > Find Packet with Display filter `http.response.code == 500`. Reopen the saved `pcapng` and confirm the comment persists.

6. Export a reproducible packet table. Record the filter, frame numbers, measurement, explanation and limitation in `outputs/findings.md`.

```
python3 scripts/export_evidence.py
```

Expected evidence

With the lab filter applied, your packet list should match the frames below (produced by TShark from this lab's own capture).

http.response.code >= 400					
No.	Time	Source	Destination	Protocol	Info
41	2.130000	192.0.2.20	192.0.2.10	HTTP	HTTP/1.1 404 Not Found (text/plain)
50	2.270000	192.0.2.20	192.0.2.10	HTTP	HTTP/1.1 500 Internal Server Error (text/plain)
branch-office.pcap Displayed: 2					

Lab 04 expected evidence

Test it

The supplied `http.response.code >= 400` expression matches 2 frame(s) in the specified capture. Run `scripts/verify.py` and compare the listed frame numbers. Keep your findings and exported table in `outputs/`. Explain the observed result rather than only copying a count.

Troubleshooting

- TShark not found: install Wireshark CLI tools and add the installation folder to PATH; on Windows use the Wireshark install directory.
- A filter returns zero: clear other filters, use the specified capture, and check the expression is in the display toolbar.

- TLS remains opaque: select the matching lab-tls.keys file by absolute path, reload, and remove a key-log preference from another lab.

Try it with TShark

The same evidence from the command line — run it from this lab folder:

```
tshark -n -r data/branch-office.pcap -Y "http.response.code >= 400" -T  
fields -e frame.number -e http.response.code -e http.request_in -e  
http.time
```

Challenge

Develop and explain an alternative filter.

Reflection

Which second observation point would strengthen your conclusion?

Extension (optional)

HTTP lab — apply your LAB HTTP Error rule to the HTTP trace from the Kurose & Ross labs. Source: J.F. Kurose and K.W. Ross, Wireshark Labs (gaia.cs.umass.edu/kurose_ross/wireshark.php).

Reset

Clear all display filters and return to the C1123-Analyst or Default profile. If you loaded a TLS key log, remove it from Preferences > Protocols > TLS. The supplied captures never need regenerating for the core lab.

Note: The same steps appear in the Learner Guide. The slides show only the scenario and a summary.